Unified Notes on Operator Cores, Trajectory Spaces, and Spectral Transport

A consolidated mathematical note on substitution, similarity, path-space prediction, and complex-valued modes

One-line takeaway

Infinity need not live in the number of states; it may live in the trajectory space, and operator cores are the laws that transport structure across representations.

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1 Scope and unifying idea

This document reorganizes several related notes into a single operator-theoretic spine. The main theme is that the same structural pattern reappears in different settings:

- in real-variable substitution, differentiation is transported across a coordinate change;
- in matrix similarity reduction, a transition operator is transported into a basis where its spectral modes are explicit;
- on the full trajectory space of a Markov chain, future observables are transported backward by conditional expectation through the transition operator;
- in the complex plane, holomorphic substitution and complex-valued predictive modes obey the same transport logic.

The compact principle is this: an *operator core* is a commutation or intertwining law that reveals what survives when one changes representation.

2 The substitution core in the real setting

Definition 2.1 (Substitution operator)**.** Let $u = u(x)$ be a smooth function, and let F be a smooth outer function. Define the substitution operator

$$U_u : \{F(u)\} \to \{G(x)\}, \qquad (U_u F)(x) := F(u(x)).$$

Let $D_x = \frac{d}{dx}$ and $D_u = \frac{d}{du}$, and define the multiplication operator

$$(M_{u'}\varphi)(x) := u'(x)\varphi(x).$$

Theorem 2.2 (Substitution core)**.** *The derivative transports through substitution according to*

$$\boxed{D_x \circ U_u = M_{u'} \circ U_u \circ D_u.}$$

Proof. For every smooth F,

$$D_x[(U_u F)(x)] = \frac{d}{dx} F(u(x)) = F'(u(x))u'(x).$$

Since $(U_u D_u F)(x) = F'(u(x))$, this becomes

$$D_x(U_u F) = M_{u'}(U_u D_u F),$$

which is exactly the claimed operator identity. □

Corollary 2.3 (Substitution rule for antiderivatives)**.** *If $F'(u) = f(u)$, then*

$$\int f(u(x))u'(x)\,dx = F(u(x)) + C.$$

Example 2.4 (Control example)**.** Take $u(x) = \sin x$ and $F(u) = u^3$. Then

$$(U_u F)(x) = \sin^3 x, \qquad D_x[\sin^3 x] = 3\sin^2 x \cos x.$$

On the other hand,

$$D_u F = 3u^2, \qquad (U_u D_u F)(x) = 3\sin^2 x, \qquad M_{u'}(U_u D_u F) = 3\sin^2 x \cos x.$$

Both sides agree.

3 Finite state spaces and infinite trajectory spaces

Definition 3.1 (Finite and infinite trajectories)**.** Let

$$S = \{0, 1\}$$

be a two-state space. Define

$$\Omega_{\text{fin}} := \bigcup_{n \geq 1} S^n, \qquad \Omega_\infty := S^{\mathbb{N}}.$$

The set Ω_{fin} is the space of finite trajectories, while Ω_∞ is the space of infinite trajectories.

Lemma 3.2. *The space Ω_{fin} is countable.*

Proof. For each fixed n, the set S^n has exactly 2^n elements and is finite. Therefore

$$\Omega_{\text{fin}} = \bigcup_{n \geq 1} S^n$$

is a countable union of finite sets, hence countable. □

Lemma 3.3. *The space $\Omega_\infty = S^{\mathbb{N}}$ has cardinality*

$$|\Omega_\infty| = 2^{\aleph_0}.$$

Proof. An element $\omega \in S^{\mathbb{N}}$ is an infinite binary sequence

$$\omega = (\omega_1, \omega_2, \omega_3, \dots), \qquad \omega_k \in \{0, 1\}.$$

Thus Ω_∞ is the set of all infinite binary sequences, whose cardinality is the cardinality of the continuum. □

Theorem 3.4. *A two-state system may have only countably many finite state histories, yet uncountably many infinite trajectories. More precisely,*

$$|\Omega_{\text{fin}}| = \aleph_0, \qquad |\Omega_\infty| = 2^{\aleph_0}.$$

Proof. The first claim follows from the first lemma and the second from the second lemma. □

Theorem 3.5 (Limits of cardinality)**.** *Cardinality alone does not determine a canonical generative interpretation of a space of objects.*

More precisely, there exist spaces A and B such that

$$|A| = |B| = 2^{\aleph_0},$$

while their natural interpretations as generated objects are different.

Proof. Take

$$A = \{0, 1\}^{\mathbb{N}}, \qquad B = \mathbb{N}^{\mathbb{N}}.$$

It is standard that

$$|\{0, 1\}^{\mathbb{N}}| = 2^{\aleph_0}, \qquad |\mathbb{N}^{\mathbb{N}}| = \aleph_0^{\aleph_0} = 2^{\aleph_0}.$$

Hence $|A| = |B|$. However, A is naturally interpreted as the space of infinite binary trajectories of a two-state system, whereas B is naturally interpreted as the space of infinite sequences of natural numbers. So equality of cardinality does not, by itself, specify a unique generative meaning. □

Corollary 3.6 (Trajectory-space takeaway)**.** *Infinity need not live in the number of states; it may already arise at the level of trajectories generated by iterating a finite transition structure.*

4 Similarity core for the two-state transition matrix

Consider the transition matrix

$$P = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}, \qquad p, q \in (0,1).$$

As an operator on $V = \mathbb{R}^2$, define

$$\mathcal{P}(v) := Pv.$$

Lemma 4.1 (Spectral data)**.** *The eigenvalues of P are*

$$\lambda_1 = 1, \qquad \lambda_2 = 1 - p - q.$$

A corresponding pair of eigenvectors is

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} p \\ -q \end{pmatrix}.$$

Proof. One checks directly that $Pv_1 = v_1$ and $Pv_2 = (1-p-q)v_2$. □

Definition 4.2 (Transport to spectral coordinates)**.** Set

$$S = \begin{pmatrix} 1 & p \\ 1 & -q \end{pmatrix}, \qquad \det S = -(p+q) \neq 0,$$

and define

$$U_S(v) := S^{-1}v, \qquad \mathcal{D}_\Lambda(z) := \Lambda z, \qquad \Lambda := S^{-1}PS = \begin{pmatrix} 1 & 0 \\ 0 & 1-p-q \end{pmatrix}.$$

Theorem 4.3 (Similarity core)**.** *The transport operator and transition operator satisfy*

$$\boxed{U_S \circ \mathcal{P} = \mathcal{D}_\Lambda \circ U_S.}$$

Equivalently,

$$\boxed{\mathcal{P} = U_S^{-1} \circ \mathcal{D}_\Lambda \circ U_S.}$$

Moreover, for every $n \geq 1$,

$$U_S \circ \mathcal{P}^n = \mathcal{D}_\Lambda^n \circ U_S, \qquad \mathcal{P}^n = U_S^{-1} \circ \mathcal{D}_\Lambda^n \circ U_S.$$

Proof. For every $v \in V$,

$$(U_S \circ \mathcal{P})(v) = S^{-1}Pv, \qquad (\mathcal{D}_\Lambda \circ U_S)(v) = \Lambda S^{-1}v.$$

Since $\Lambda = S^{-1}PS$, one has $\Lambda S^{-1} = S^{-1}P$, so both sides agree. The n-step formula follows by iteration. □

Example 4.4 (Explicit n-step evolution)**.** Let

$$x_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Then

$$S^{-1} = \frac{1}{p+q}\begin{pmatrix} q & p \\ 1 & -1 \end{pmatrix}, \qquad z_0 := U_S(x_0) = \frac{1}{p+q}\begin{pmatrix} q \\ 1 \end{pmatrix}.$$

After n steps,

$$z_n = \mathcal{D}_\Lambda^n(z_0) = \frac{1}{p+q}\begin{pmatrix} q \\ (1-p-q)^n \end{pmatrix}.$$

Transforming back,

$$x_n = Sz_n = \frac{1}{p+q}\begin{pmatrix} q + p(1-p-q)^n \\ q - q(1-p-q)^n \end{pmatrix}.$$

So the operator splits into a stationary mode and a decaying mode.

5 Operator core for trajectories on the full path space

Path space and lifted observables

Let $\Omega = S^{\mathbb{N}}$ and let

$$X_n : \Omega \to S, \qquad X_n(\omega) = \omega_n$$

denote the n-th coordinate process.

View P as an operator on observables $f : S \to \mathbb{C}$ via

$$(Pf)(i) := \sum_{j \in S} P_{ij} f(j).$$

For each $n \geq 0$, define the lifting operator

$$U_n : \{f : S \to \mathbb{C}\} \to \{F : \Omega \to \mathbb{C}\}, \qquad (U_n f)(\omega) := f(X_n(\omega)).$$

Let

$$\mathcal{F}_n := \sigma(X_0, \ldots, X_n)$$

be the natural filtration, and define the conditional expectation operator

$$\mathbb{E}_n(F) := \mathbb{E}[F \mid \mathcal{F}_n].$$

Theorem 5.1 (Trajectory core)**.** *For the coordinate process of the Markov chain with transition operator P,*

$$\boxed{\mathbb{E}_n \circ U_{n+1} = U_n \circ P.}$$

More generally, for every $k \geq 1$,

$$\boxed{\mathbb{E}_n \circ U_{n+k} = U_n \circ P^k.}$$

Equivalently, for every observable $f : S \to \mathbb{C}$,

$$\mathbb{E}[f(X_{n+1}) \mid \mathcal{F}_n] = (Pf)(X_n),$$

and

$$\mathbb{E}[f(X_{n+k}) \mid \mathcal{F}_n] = (P^k f)(X_n).$$

Proof. By the Markov property,

$$\mathbb{E}[f(X_{n+1}) \mid X_0, \ldots, X_n] = \mathbb{E}[f(X_{n+1}) \mid X_n].$$

If $X_n = i$, then by definition of P,

$$\mathbb{E}[f(X_{n+1}) \mid X_n = i] = \sum_{j \in S} P_{ij} f(j) = (Pf)(i).$$

Therefore

$$\mathbb{E}[f(X_{n+1}) \mid \mathcal{F}_n] = (Pf)(X_n),$$

which is precisely $\mathbb{E}_n(U_{n+1}f) = U_n(Pf)$. The k-step version follows from the same argument with P^k. □

Remark 5.2 (Meaning of the trajectory core)**.** This is the path-space analogue of the substitution core. In the substitution case, a derivative is transported across a coordinate change. In the trajectory case, a future observable is transported backward to the present by prediction through P.

Corollary 5.3 (Trajectory spectral law)**.** *If $\phi : S \to \mathbb{C}$ is an eigenobservable of P, so that*

$$P\phi = \lambda\phi,$$

and we define the lifted mode

$$Z_n(\omega) := \phi(X_n(\omega)),$$

then

$$\boxed{\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \lambda Z_n.}$$

More generally,

$$\mathbb{E}[Z_{n+k} \mid \mathcal{F}_n] = \lambda^k Z_n.$$

Thus eigenmodes of the finite-state operator become predictive modes on the full path space.

Example 5.4 (Present prediction of a future event)**.** Take the observable

$$f(0) = 1, \qquad f(1) = 0.$$

Then $U_n f = \mathbf{1}_{\{X_n=0\}}$. The trajectory core gives

$$\mathbb{E}[\mathbf{1}_{\{X_{n+1}=0\}} \mid \mathcal{F}_n] = (Pf)(X_n).$$

Since

$$(Pf)(0) = 1 - p, \qquad (Pf)(1) = q,$$

we obtain

$$\mathbb{E}[\mathbf{1}_{\{X_{n+1}=0\}} \mid \mathcal{F}_n] = \begin{cases} 1-p, & X_n = 0, \\ q, & X_n = 1. \end{cases}$$

So the future event "being in state 0" becomes a present measurable predictor on the full path space.

6 Complex plane: substitution core and complex-valued trajectory modes

Definition 6.1 (Holomorphic substitution operator)**.** Let $u = u(z)$ be holomorphic on a domain $\Omega \subset \mathbb{C}$, and let F be holomorphic on a domain containing $u(\Omega)$. Define

$$U_u : H(u(\Omega)) \to H(\Omega), \qquad (U_u F)(z) := F(u(z)).$$

Let $D_z = \frac{d}{dz}$, $D_w = \frac{d}{dw}$, and define

$$(M_{u'}\varphi)(z) := u'(z)\varphi(z).$$

Theorem 6.2 (Complex substitution core)**.** *The holomorphic substitution operator satisfies*

$$\boxed{D_z \circ U_u = M_{u'} \circ U_u \circ D_w.}$$

Proof. For every holomorphic F,

$$D_z[(U_u F)(z)] = \frac{d}{dz} F(u(z)) = F'(u(z))u'(z).$$

Since $(U_u D_w F)(z) = F'(u(z))$, the result follows. □

Corollary 6.3 (Complex substitution in integrals)**.** *If $F'(w) = f(w)$, then*

$$\int f(u(z))u'(z)\,dz = F(u(z)) + C.$$

Along a piecewise smooth curve γ,

$$\int_\gamma f(u(z))u'(z)\,dz = \int_{u\circ\gamma} f(w)\,dw.$$

Example 6.4 (Complex control example)**.** Take

$$u(z) = z^2, \qquad F(w) = w^3.$$

Then

$$(U_u F)(z) = F(z^2) = z^6.$$

The left-hand side is

$$D_z(U_u F)(z) = \frac{d}{dz} z^6 = 6z^5.$$

The right-hand side is

$$D_w F(w) = 3w^2, \qquad (U_u D_w F)(z) = 3z^4, \qquad u'(z) = 2z,$$

so

$$(M_{u'} \circ U_u \circ D_w)(F)(z) = 2z \cdot 3z^4 = 6z^5.$$

Both sides agree.

Remark 6.5 (Complex-valued trajectory modes)**.** The trajectory core remains valid for observables $\phi : S \to \mathbb{C}$. If

$$P\phi = \lambda\phi, \qquad \lambda \in \mathbb{C},$$

and

$$Z_n := \phi(X_n),$$

then

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \lambda Z_n.$$

Writing $\lambda = re^{i\theta}$, one sees that the predictive mode evolves by magnitude scaling r and phase rotation θ:

$$\boxed{\mathbb{E}_n Z_{n+1} = re^{i\theta} Z_n.}$$

This is the path-space analogue of a damped rotating mode.

7 Bridge-lock and quartic excess

Lemma 7.1 (Uniform quartic window lower bound)**.** *Let*

$$\delta_\rho(u) = au + b(u), \qquad a \neq 0, \qquad |b(u)| \leq M \quad \forall u \in \mathbb{R},$$

and define

$$P_4(\rho; \tau, L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du.$$

Then for every

$$\tau \geq \tau_0 := \frac{2M}{|a|} \qquad \text{and every } L > 0,$$

one has

$$P_4(\rho; \tau, L) \geq \frac{|a|^4}{16} \int_\tau^{\tau+L} u^4 \, du,$$

hence

$$P_4(\rho; \tau, L) \geq \frac{|a|^4}{16} \left(\tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5 \right).$$

Proof. Since $|b(u)| \leq M$, the reverse triangle inequality gives

$$|\delta_\rho(u)| = |au + b(u)| \geq |a|u - M.$$

If $u \geq \tau \geq 2M/|a|$, then $|a|u - M \geq |a|u/2$, hence

$$|\delta_\rho(u)|^4 \geq \frac{|a|^4}{16} u^4.$$

Integrating yields the claim. □

Theorem 7.2 (Explosive excess from a linear drift)**.** *Let*

$$E_{\text{ex}}(f) := \limsup_{L \to \infty} \sup_{\tau \in \mathbb{R}} \frac{Q_{\text{bank}}(f; \tau, L) - E_0 L}{L}.$$

Assume that for some off-critical component ρ,

$$\delta_\rho(u) = au + b(u), \qquad a \neq 0, \qquad |b(u)| \leq M \quad \forall u,$$

and define

$$P_4(\rho; \tau, L) := \int_\tau^{\tau+L} |\delta_\rho(u)|^4 \, du.$$

Assume moreover that there exist constants $c_1 > 0$, $c_2 \geq 0$, $\tau_1 \in \mathbb{R}$, $L_1 \geq 1$ such that for all $\tau \geq \tau_1$ and $L \geq L_1$,

$$Q_{\text{bank}}(f; \tau, L) - E_0 L \geq c_1 P_4(\rho; \tau, L) - c_2 L.$$

Then

$$E_{\text{ex}}(f) = +\infty.$$

Proof. Choose

$$\tau \geq \tau_* := \max\left\{ \tau_1, \frac{2M}{|a|} \right\}.$$

By the quartic lower bound,

$$P_4(\rho; \tau, L) \geq \frac{|a|^4}{16} \left(\tau^4 L + 2\tau^3 L^2 + 2\tau^2 L^3 + \tau L^4 + \frac{1}{5} L^5 \right).$$

Substituting into the bridge inequality and dividing by L gives

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \geq \frac{c_1|a|^4}{16}\left(\tau^4 + 2\tau^3 L + 2\tau^2 L^2 + \tau L^3 + \frac{1}{5}L^4\right) - c_2.$$

The right-hand side contains the positive quartic term

$$\frac{c_1|a|^4}{80}L^4,$$

so it diverges to $+\infty$ as $L \to \infty$. Hence

$$E_{\text{ex}}(f) = +\infty.$$

□

Remark 7.3 (Soft phase reading)**.** A useful interpretation is that a sharp sign-collapse point may be replaced by a regularized phase interval while the magnitude channel survives. In that reading, the quartic drift does not disappear across the phase change, and the bridge-lock mechanism remains active through the singular region.

8 Synthesis

The notes above fit into one shared template. In each case there is:

1. a space of objects or observables,
2. a transport operator that changes representation,
3. a structural law that survives the change.

For quick reference:

Setting	**Transport**	**Core law**
Real substitution	U_u	$D_x \circ U_u = M_{u'} \circ U_u \circ D_u$
Matrix similarity	U_S	$U_S \circ \mathcal{P} = \mathcal{D}_\Lambda \circ U_S$
Trajectory space	U_n with $\mathbb{E}_n$	$\mathbb{E}_n \circ U_{n+1} = U_n \circ P$
Complex substitution	U_u	$D_z \circ U_u = M_{u'} \circ U_u \circ D_w$
Complex trajectory modes	lifted eigenobservable Z_n	$\mathbb{E}_n Z_{n+1} = \lambda Z_n$

The deep point is that the state space may be finite while the generated trajectory space is uncountable, and the relevant asymptotic information is carried not by raw coordinates but by the operator cores that transport structure across viewpoints.

Final takeaway

Infinity need not live in the number of states. It may live in the trajectory space, while operator cores are the laws that reveal what survives under transport, prediction, and spectral reduction.